DSE Conformance Report

Velexa matching engine vs the order types the OMS sends to DSE · generated 13 October 2026

Scope. 184 cases were driven through the engine's real FIX acceptor (FIXT.1.1 / FIX 5.0 SP2, strict data dictionary on) using the exact tag set listed in DSE FIX Order Types · Velexa OMS (11 Oct 2026). Expected DSE behaviour comes from that document, its observed DSE rejects (price band ±7.5% / ±8.75%, minimum quantity step 1000) and the FIX specification. Where DSE behaviour is not specified the case is marked Info and not judged. No engine code was changed. Reproduce with `go test -run TestConformance` then `python3 tools/conformance_report.py`.

184	122	6	44	12
cases run	match DSE	deviate (blockers)	gaps	partial / info

Headline findings

Finding	Severity	Detail
Orders that DSE would refuse are accepted OT-02, OT-095/099, SD-4	Deviates	A limit order with no price executes as a market order; unknown OrdTypes (5, 9) become limit orders; a Sell-Short (54=5) executes as a **buy**.
Fill reports are not valid FIX ER-2, CM-3	Deviates	Every fill carries ExecType 150=TRADE (FIX requires F). The Order Cancel Reject (35=9) omits the required OrderID (37).
OMS features are not understood by the engine OT-06, TIF-S, EI-2…6, QO-1, QO-3, QO-4	Gap	Market-At-Best (Z), Validity S, ExecInst n/S/q/w (Park / Unpark / Block) are refused by the dictionary or ignored. DisplayQty (1138) and MinQty (110) have no effect — the engine only reads MaxFloor (111).
Board groups (762) are never validated BG-02 … BG-17	Gap	All 16 non-PUBLIC groups are accepted on a MAIN-board equity (including INDEX, ATBX, SCPX, debt and block groups). Tag 762 is never read.
Order Status Request (35=H) is ignored CM-11, CM-12	Gap	No reply of any kind, so the OMS cannot reconcile order state against the exchange.
Amend (35=G) skips pre-trade checks CM-7, CM-8	Gap	Amending a resting order to an off-tick price or a price far outside the circuit band is accepted.
GTD / time-in-force semantics TIF-6a, TIF-6b, TIF-2/5/7, ST-1	Gap	GTD without an expiry never expires; an expiring order is removed silently with no Execution Report; At-Open / GTX / At-Close are treated as Day; Day orders never expire at close.
Auction fills are not reported PH-3, PH-4	Gap	The Pre-Open uncross trades orders but sends no Execution Report to the broker; Pre-Open also skips the tick and price-band checks.

What works: limit and market matching with price-time priority, partial fills, IOC and FOK, cancel and amend of resting orders, the full DSE price-band ladder (all 64 boundary probes correct), tick-size enforcement, halted and unknown symbols, whole-share quantities, the per-instrument market lot (minimum quantity step,

set from the Instruments page), required-field validation, Pre-Open handling of market orders, and the session messages Logon / Logout / TestRequest / ResendRequest / SequenceReset.

Results by section

Section	Cases	Match	Deviate	Gap	Partial / info
Order type (40)	10	3	3	1	3
Validity (59)	14	8	0	6	0
Side & quantity	13	11	1	1	0
Board group (762)	20	4	0	16	0
ExecInst (18)	6	0	0	4	2
Quantity options	4	0	0	3	1
Tags & reports	14	9	1	2	2
Price band / tick	70	69	0	0	1
Cancel / modify / status	13	6	1	5	1
Phases	5	3	0	2	0
Matching	3	3	0	0	0
Rules not modelled	6	0	0	4	2
Session messages	6	6	0	0	0

Comparison: OMS / DSE vs engine

1. Order type (40)

ID	Case	DSE / OMS expects	Matching engine did	Result
OT-01	Limit (40=2) with price	Accepted, rests on book	`35=8 150=0 39=0`	Matches DSE
OT-02	Limit (40=2) without price (44), liquid book	Rejected: price required for a limit order	`35=8 150=TRADE 39=2 14=100` accepted and executed — the engine silently converts a 40=2 order with no price into a MARKET order (fixacceptor.go:465-479)	Deviates
OT-03	Market (40=1) without price, liquid book	Matches at best available price	`35=8 150=TRADE 39=2 14=100`	Matches DSE
OT-04	Market (40=1) carrying a price (post-close style)	Accepted; DSE requires the close price in the post-close session	`35=8 150=TRADE 39=2 14=100`	Matches DSE
OT-05	Market (40=1) on an empty book	Order not executed (no liquidity); treatment of the remainder not specified by the OMS reference	`35=8 150=8 39=8 58=no liquidity: no resting SELL orders on T004M` rejected with 'no liquidity' — market orders never rest	Info
OT-06	Market-At-Best (40=Z)	DSE-specific type: accepted and matched at best	`35=3 371=40 372=D 373=5 58=Value is incorrect (out of range) for this tag` OrdType Z is not in the engine's data dictionary, so the session layer refuses it (35=3) — Market-At-Best is not supported	Gap
OT-07	Stop (40=3) with StopPx (99)	Not used by the OMS (informational)	`35=8 150=A 39=A` engine: 35=8 150=A 39=A	Info
OT-08	Stop-limit (40=4) with StopPx and price	Not used by the OMS (informational)	`35=8 150=A 39=A` engine: 35=8 150=A 39=A	Info
OT-095	Unsupported OrdType 40=5	Rejected as an unsupported order type	`35=8 150=0 39=0` accepted and silently treated as a Limit order (mapping default in fixacceptor.go:465-479)	Deviates
OT-099	Unsupported OrdType 40=9	Rejected as an unsupported order type	`35=8 150=0 39=0` accepted and silently treated as a Limit order (mapping default in fixacceptor.go:465-479)	Deviates

2. Validity (59)

ID	Case	DSE / OMS expects	Matching engine did	Result
TIF-0	Day (59=0) resting limit	Accepted, rests until close	`35=8 150=0 39=0`	Matches DSE

ID	Case	DSE / OMS expects	Matching engine did	Result
TIF-1	GTC (59=1) resting limit	Accepted, rests until cancelled	`35=8 150=0 39=0`	Matches DSE
TIF-3a	IOC (59=3) partially fillable (50 of 100)	Fills 50, cancels the remaining 50	`35=8 150=TRADE 39=1 14=50`	Matches DSE
TIF-3b	IOC (59=3) with no liquidity	Cancelled / expired with no fill	`35=8 150=4 39=4`	Matches DSE
TIF-4a	FOK (59=4) fully fillable	Fully filled	`35=8 150=TRADE 39=2 14=100`	Matches DSE
TIF-4b	FOK (59=4) insufficient liquidity	Whole order cancelled, no partial fill	`35=8 150=4 39=4`	Matches DSE
TIF-6a	GTD (59=6) without expiry date (OMS sends none)	Rejected without ExpireDate(432)/ ExpireTime(126) — OMS reference says DSE 'will probably reject'	`35=8 150=0 39=0` accepted; with no expiry the order never expires (parseExpire returns zero)	Gap
TIF-6b	GTD (59=6) with ExpireTime (126) 3 s ahead — expires	Order removed at expiry AND an Execution Report (150=C Expired) is sent	`35=8 150=0 39=0` order removed from the book at expiry but NO Execution Report (150=C) was sent — the broker is never told	Gap
TIF-6c	GTD (59=6) with ExpireDate (432) only	Accepted; expires at end of that date	`35=8 150=0 39=0`	Matches DSE
TIF-2	At the Opening (59=2)	Honoured as its own validity (not offered by any OMS screen, but the builder can pass it)	`35=8 150=0 39=0` accepted but silently treated as Day (mapTIF default) — semantics not implemented	Gap
TIF-5	Good Till Crossing (59=5)	Honoured as its own validity (not offered by any OMS screen, but the builder can pass it)	`35=8 150=0 39=0` accepted but silently treated as Day (mapTIF default) — semantics not implemented	Gap
TIF-7	At the Close (59=7)	Honoured as its own validity (not offered by any OMS screen, but the builder can pass it)	`35=8 150=0 39=0` accepted but silently treated as Day (mapTIF default) — semantics not implemented	Gap
TIF-S	Session (59=S) — offered by the OMS trader UI	Accepted (the OMS offers it); not a FIX 5.0 SP1/SP2 enum value	`35=3 371=59 372=D 373=5 58=Value is incorrect (out of range) for this tag` rejected by the strict dictionary (35=3) — an order the OMS UI allows can never reach the engine	Gap
TIF-X	Missing TimeInForce	Defaults to Day	`35=8 150=0 39=0`	Matches DSE

3. Side & quantity

ID	Case	DSE / OMS expects	Matching engine did	Result
SD-1	Buy (54=1)	Accepted	`35=8 150=0 39=0`	Matches DSE

ID	Case	DSE / OMS expects	Matching engine did	Result
SD-2	Sell (54=2)	Accepted	`35=8 150=0 39=0`	Matches DSE
SD-3	Missing Side (54)	Rejected	`35=3 371=54 372=D 373=1 58=Required tag missing`	Matches DSE
SD-4	Side 54=5 (sell short)	Rejected (the OMS reference: no short-sell side)	`35=8 150=0 39=0` accepted — any Side other than '2' is treated as a BUY (fixacceptor.go:461-464), so a sell-short would execute as a purchase	Deviates
QT-1	OrderQty = 0	Rejected	`35=8 150=8 39=8 58=invalid OrderQty (tag 38): 0 — quantity must be greater than zero`	Matches DSE
QT-2	Fractional OrderQty (10.5)	Rejected (whole shares only)	`35=8 150=8 39=8 58=invalid OrderQty (tag 38): 10.5 — the exchange trades whole shares only`	Matches DSE
QT-3	Negative OrderQty	Rejected	`35=8 150=8 39=8 58=invalid OrderQty (tag 38): -5 — quantity must be greater than zero`	Matches DSE
QT-4	Quantity a multiple of 1000	Accepted	`35=8 150=0 39=0`	Matches DSE
QT-5	Quantity 1500 on a lot-1000 instrument	Rejected: '(792) Minimum quantity step: 1000' (observed DSE reject, instrument-dependent)	`35=8 150=8 39=8 58=Minimum quantity step: 1000`	Matches DSE
QT-5b	Quantity 2000 on a lot-1000 instrument	Accepted (whole lots)	`35=8 150=0 39=0`	Matches DSE
QT-6	Duplicate ClOrdID (11) resubmitted	Rejected as a duplicate order	`35=8 150=0 39=0` second order with the same ClOrdID accepted — no duplicate-ClOrdID check (also breaks cancel/replace by ClOrdID)	Gap
QT-7	Missing Symbol (55)	Rejected	`35=8 150=8 39=8 58=missing Symbol (tag 55)`	Matches DSE
QT-8	Unknown symbol	Rejected (not in the security master)	`35=8 150=8 39=8 58=unknown symbol NOSUCHSYM: not in this engine's instrument universe (unknown)`	Matches DSE

4. Board group (762)

ID	Case	DSE / OMS expects	Matching engine did	Result
BG-01	762=PUBLIC on a MAIN-board equity	Accepted (public market, MAIN board)	`35=8 150=0 39=0`	Matches DSE
BG-02	762=ATBPUB on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap

ID	Case	DSE / OMS expects	Matching engine did	Result
BG-03	762=SPUBLIC on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-04	762=BLOCK on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-05	762=SBLOCK on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-06	762=BUYIN on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-07	762=ATBBUYIN on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-08	762=SBUYIN on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-09	762=DEBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-10	762=ATBDEBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-11	762=SDEBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-12	762=ALTDBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-13	762=BUYDBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-14	762=YIELDDBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap

ID	Case	DSE / OMS expects	Matching engine did	Result
BG-15	762=INDEX on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-16	762=ATBX on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-17	762=SCPX on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-18	762=SPUBLIC on an SME-board equity	Accepted	`35=8 150=0 39=0`	Matches DSE
BG-19	762=ATBPUB on an ATB-board equity	Accepted	`35=8 150=0 39=0`	Matches DSE
BG-20	Missing 762	Accepted (defaults to the instrument's group)	`35=8 150=0 39=0`	Matches DSE

5. ExecInst (18)

ID	Case	DSE / OMS expects	Matching engine did	Result
EI-1	18=Placed (n) on a new order	Normal placed order — rests on the book	`35=8 150=0 39=0` accepted; instruction not interpreted (tag 18 never read)	Partial
EI-2	18=Park (S) on a new order	Order parked: held off the book, no matching	`35=8 150=0 39=0` accepted but the order rests on the book — Park is not implemented (tag 18 never read)	Gap
EI-3	18=Block (w) on a new order	Block-board order	`35=3 371=18 372=D 373=5 58=Value is incorrect (out of range) for this tag` rejected — 'w' is not in the engine's tag-18 dictionary enum, so the OMS's DSE-specific instruction never reaches the engine	Gap
EI-4	18=Block Park (w S) on a new order	Block order parked	`35=3 371=18 372=D 373=5 58=Value is incorrect (out of range) for this tag` rejected — 'w S' is not in the engine's tag-18 dictionary enum, so the OMS's DSE-specific instruction never reaches the engine	Gap
EI-5	18=q (Unpark) on a modify (35=G)	Un-parks the order and places it on the book	`35=8 150=5 39=0` accepted as an ordinary amend; instruction ignored	Partial
EI-6	18=w q (Unpark) on a modify (35=G)	Un-parks the order and places it on the book	`35=3 371=18 372=G 373=5 58=Value is incorrect (out of range) for this tag` rejected — Unpark ('w q') cannot be sent through the dictionary-validated session	Gap

6. Quantity options

ID	Case	DSE / OMS expects	Matching engine did	Result
Q0-1	DisplayQty (1138=30) iceberg on a 100-lot (the OMS's tag)	Only 30 shown on the book	`35=8 150=0 39=0` book shows 100 — tag 1138 is ignored (the engine reads tag 111 MaxFloor instead), so the iceberg is not applied to OMS orders	Gap
Q0-2	MaxFloor (111=30) iceberg (engine's own tag)	(not sent by the OMS) 30 shown	`35=8 150=0 39=0` 111 honoured — book shows 30	Info
Q0-3	MinQty (110) greater than OrderQty (500 > 100)	Rejected	`35=8 150=0 39=0` accepted — MinQty (110) is ignored	Gap
Q0-4	MinQty (110=50) with only 30 available	No fill (minimum quantity cannot be met)	`35=8 150=TRADE 39=1 14=30` filled below the minimum quantity — MinQty (110) is ignored	Gap

7. Tags & reports

ID	Case	DSE / OMS expects	Matching engine did	Result
TG-1	Party group 453 (448/447/452) present	Accepted	`35=8 150=0 39=0`	Matches DSE
TG-2	No party group	Accepted	`35=8 150=0 39=0`	Matches DSE
TG-3	OrderRestrictions 529=N (non-standard)	Accepted (DSE accepted it in logged traffic)	`35=8 150=0 39=0`	Matches DSE
TG-4	Missing Account (1)	Accepted or rejected per DSE (BO ID required by the OMS)	`35=8 150=0 39=0` engine: 35=8 150=0 39=0	Info
TG-5	Missing TransactTime (60)	Rejected (required field)	`35=3 371=60 372=D 373=1 58=Required tag missing`	Matches DSE
TG-6	Missing ClOrdID (11)	Rejected	`35=3 371=11 372=D 373=1 58=Required tag missing`	Matches DSE
ER-1	Fill ER carries FIX-required tags (37,17,150,39,55,54,151,14,11)	All present	`35=8 150=TRADE 39=2 14=100`	Matches DSE
ER-2	Fill ER ExecType (150)	150=F (Trade)	`35=8 150=TRADE 39=2 14=100` 150=TRADE — not a valid FIX 5.0 value (must be 'F')	Deviates
ER-3	ER echoes Account (1)	Echoed (OMS correlates by BO ID)	`35=8 150=TRADE 39=2 14=100` tag 1 not echoed on the execution report	Gap
ER-4	ER echoes OrdType (40) / TimeInForce (59)	Echoed	`35=8 150=TRADE 39=2 14=100` tags 40/59 not echoed	Gap
ER-5	ER carries last fill price/qty (31/32)	31 and 32 on a fill	`35=8 150=TRADE 39=2 14=100`	Matches DSE

ID	Case	DSE / OMS expects	Matching engine did	Result
ER-6	ER carries average price (6) and cumulative qty (14)	Both present	`35=8 150=TRADE 39=2 14=100`	Matches DSE
ER-7	ER carries security id (48/22) and board	Instrument identification present	`35=8 150=TRADE 39=2 14=100`	Matches DSE
ER-8	Reject ER carries OrdRejReason (103) and Text (58)	103 + 58 on every rejected order	`35=8 150=8 39=8 58=price band: 50.00 outside +/-10.00% limit [90.00-110.00] of prev 100.00` Text (58) present but OrdRejReason (103) is never set	Partial

8. Price band / tick

ID	Case	DSE / OMS expects	Matching engine did	Result
PT-1	Price on tick (100.10)	Accepted	`35=8 150=0 39=0`	Matches DSE
PT-2	Price off tick (100.05)	Rejected: off-tick	`35=8 150=8 39=8 58=off-tick: 100.0500 is not a multiple of tick 0.10`	Matches DSE
PT-3	Price 100.02 (sub-paisa tick)	Rejected: off-tick	`35=8 150=8 39=8 58=off-tick: 100.0200 is not a multiple of tick 0.10`	Matches DSE
PT-4	Market order ignores the price band	DSE applies circuit limits to execution prices	`35=8 150=TRADE 39=2 14=100` market orders skip band/tick checks (engine.go:297 applies them to limits only): 35=8 150=TRADE 39=2 14=100	Info
PT-5	Order on a halted instrument	Rejected	`35=8 150=8 39=8 58=trading halted in T045M`	Matches DSE
PT-6	Order after the halt is lifted	Accepted	`35=8 150=0 39=0`	Matches DSE

DSE circuit-limit tiers — boundary test (64/64 correct)

Each tier is probed at the last accepted price and the next tick beyond it, both sides. Tiers: ±10% to Tk200, ±8.75% to Tk500, ±7.5% to Tk1000, ±6.25% to Tk2000, ±5% to Tk5000, ±3.75% above.

Previous close	Band	At upper limit	Just above	At lower limit	Just below
Tk 100.00	±10.00%	110.00 ✔	110.10 ✔	90.00 ✔	89.90 ✔
Tk 200.00	±10.00%	220.00 ✔	220.10 ✔	180.00 ✔	179.90 ✔
Tk 200.10	±8.75%	217.60 ✔	217.70 ✔	182.60 ✔	182.50 ✔
Tk 300.00	±8.75%	326.20 ✔	326.30 ✔	273.80 ✔	273.70 ✔
Tk 500.00	±8.75%	543.70 ✔	543.80 ✔	456.30 ✔	456.20 ✔
Tk 500.10	±7.50%	537.60 ✔	537.70 ✔	462.60 ✔	462.50 ✔
Tk 800.00	±7.50%	860.00 ✔	860.10 ✔	740.00 ✔	739.90 ✔
Tk 1000.00	±7.50%	1075.00 ✔	1075.10 ✔	925.00 ✔	924.90 ✔

Previous close	Band	At upper limit	Just above	At lower limit	Just below
Tk 1000.10	±6.25%	1062.60 ✔	1062.70 ✔	937.60 ✔	937.50 ✔
Tk 1500.00	±6.25%	1593.70 ✔	1593.80 ✔	1406.30 ✔	1406.20 ✔
Tk 2000.00	±6.25%	2125.00 ✔	2125.10 ✔	1875.00 ✔	1874.90 ✔
Tk 2000.10	±5.00%	2100.10 ✔	2100.20 ✔	1900.10 ✔	1900.00 ✔
Tk 3000.00	±5.00%	3150.00 ✔	3150.10 ✔	2850.00 ✔	2849.90 ✔
Tk 5000.00	±5.00%	5250.00 ✔	5250.10 ✔	4750.00 ✔	4749.90 ✔
Tk 5000.10	±3.75%	5187.60 ✔	5187.70 ✔	4812.60 ✔	4812.50 ✔
Tk 8000.00	±3.75%	8300.00 ✔	8300.10 ✔	7700.00 ✔	7699.90 ✔

9. Cancel / modify / status

ID	Case	DSE / OMS expects	Matching engine did	Result
CM-1	Cancel a resting order (35=F)	Execution Report 150=4 Cancelled	`35=8 150=4 39=4`	Matches DSE
CM-2	Cancel an unknown order	Order Cancel Reject (35=9)	`35=9 58=unknown or already-done order 102=1`	Matches DSE
CM-3	35=9 carries OrderID (37) — required by FIX	37 present on the reject	`35=9 58=unknown or already-done order 102=1` 35=9 has no OrderID (37) — fixacceptor.go:531-543	Deviates
CM-4	Modify price WITHOUT Symbol/762 (as the OMS sends 35=G)	Execution Report 150=5 Replaced	`35=8 150=5 39=0`	Matches DSE
CM-5	Modify quantity 100 -> 60	150=5 Replaced with the new quantity	`35=8 150=5 39=0`	Matches DSE
CM-6	Modify an unknown order	35=9	`35=9 58=unknown or already-done order 102=1`	Matches DSE
CM-7	Modify to a price off tick (96.05)	Rejected: off-tick	`35=8 150=5 39=0` accepted — replace does not re-run the tick check	Gap
CM-8	Modify to a price outside the band (150.00)	Rejected: outside price band	`35=8 150=5 39=0` accepted — replace does not re-run the price-band check, so an order can be amended to any price	Gap
CM-9	Modify using the latest ClOrdID chain (replace a replaced order)	Accepted	`35=8 150=5 39=0`	Matches DSE
CM-10	Replace ER echoes OrigClOrdID (41)	41 echoed on the replace/cancel ER	`35=8 150=5 39=0` OrigClOrdID (41) not echoed on the execution report	Gap
CM-11	Order Status Request (35=H) on a live order	Execution Report with the current status (150=I)	`no response` no reply — 35=H is not handled (fixacceptor.go:403-410); the OMS cannot reconcile order state	Gap

ID	Case	DSE / OMS expects	Matching engine did	Result
CM-12	Order Status Request (35=H) on an unknown order	Reject (35=8 150=8 or 35=j)	`no response` no reply — 35=H is not handled	Gap
CM-13	Modify a parked stop order	(not used by the OMS) Replaced	`35=9 58=unknown or already-done order 102=1` engine: 35=9 58=unknown or already-done order 102=1	Info

10. Phases

ID	Case	DSE / OMS expects	Matching engine did	Result
PH-1	Limit order during Pre-Open	Accepted into the call auction, no matching	`35=8 150=0 39=0`	Matches DSE
PH-2	Market order during Pre-Open	Rejected (no market orders in the auction)	`35=8 150=8 39=8 58=market orders are not accepted during the Pre-Open call auction — send a limit order`	Matches DSE
PH-3	Limit far outside the band during Pre-Open	Rejected — circuit limits apply in every session	`35=8 150=0 39=0` accepted — Pre-Open skips the tick and price-band checks (engine.go:278-289 returns before them)	Gap
PH-4	Uncross at the Pre-Open -> Continuous transition	Matching orders trade at the equilibrium price	`no response` the crossed order was executed in the uncross but the broker received NO Execution Report — auction fills are not reported over FIX	Gap
PH-5	GTC order survives a phase change	Still on the book	`no response`	Matches DSE

11. Matching

ID	Case	DSE / OMS expects	Matching engine did	Result
MT-1	Buy limit crosses the best ask — price improvement	Executes at the resting (better) price 100.00	`35=8 150=TRADE 39=2 14=100`	Matches DSE
MT-2	Price-time priority (two asks at the same price)	Earlier order fills first	`35=8 150=TRADE 39=2 14=100`	Matches DSE
MT-3	Partial fill then rest (day limit)	Fills 40, rests the remaining 60	`35=8 150=TRADE 39=1 14=40`	Matches DSE

12. Rules not modelled

ID	Case	DSE / OMS expects	Matching engine did	Result
ST-1	Day orders expire at market close	Day orders are removed at close and reported	`engine.go:35 — 'this continuous sim has no EOD cutoff': Day and GTC behave identically; nothing purges Day orders` no end-of-day processing	Gap
ST-2	Post-close session: Market order priced at the close price	DSE requires the close price on market orders after close	`engine has only the phases 'Pre-Open' and 'Continuous Trading' (engine.go:161, 627-641); no post-close/close-price handling` phase not modelled	Gap
ST-3	Automatic circuit breaker / volatility halt	Instrument halts on breaching limits; DSE halts per board rules	`only a static price-band check plus the manual per-symbol halt added in the admin console; no automatic trigger` not modelled	Gap
ST-4	Category-based rules (A/B/G/N/Z) and board-specific rules	Different lot sizes / bands / sessions by category and board	`Instrument has Board/ISIN/Tick only; no category, market lot, face value; band tiers are board-independent` not modelled	Gap
ST-5	Pre-trade RMS (buying power, holdings, position limit) on the FIX path	OMS-side check; the exchange simulator should also enforce	`rms.go RiskEngine.Check is called only from bench.go:352, never from the FIX acceptor` OMS performs these checks; the ME does not duplicate them	Info
ST-6	Order-status / execution-report replay on reconnect	Missed ERs recoverable (ResendRequest honoured)	`in-memory message store only (quickfix.NewMemoryStoreFactory) with ResetOnLogon=Y — nothing survives a reconnect` by design for a simulator	Info

13. Session messages

ID	Case	DSE / OMS expects	Matching engine did	Result
SS-1	Logon (35=A) with ResetSeqNumFlag 141=Y	Session established	`initiator sent 141=Y (ResetOnLogon=Y); session reached logged-on state`	Matches DSE
SS-2	TestRequest (35=1, 112=PING)	Heartbeat (35=0) echoing 112	`35=0`	Matches DSE
SS-3	ResendRequest (35=2, 7=1, 16=0)	Missed messages resent or gap-filled	`35=(log) ; 35=(log) ; 35=(log) ; 35=(log)`	Matches DSE
SS-4	SequenceReset (35=4, 123=N, 36=5000)	Accepted; peer's expected sequence moves	`no response`	Matches DSE
SS-5	Logout (35=5)	Logout acknowledged, session closed	`35=5`	Matches DSE
SS-6	Heartbeat (35=0) interval 30 s	Sent automatically	`handled by the QuickFIX session layer (HeartBtInt negotiated at logon)`	Matches DSE

Checklist

☑ matches DSE · ☐ needs work (Blocker = engine accepts/rejects the opposite of DSE or emits non-FIX output; Gap = feature missing or silently ignored).

Order type (40)

- ☑ `OT-01` Limit (40=2) with price
- ☐ `OT-02` Limit (40=2) without price (44), liquid book **Blocker**
- ☑ `OT-03` Market (40=1) without price, liquid book
- ☑ `OT-04` Market (40=1) carrying a price (post-close style)
- ☐ `OT-05` Market (40=1) on an empty book **Info**
- ☐ `OT-06` Market-At-Best (40=Z) **Gap**
- ☐ `OT-07` Stop (40=3) with StopPx (99) **Info**
- ☐ `OT-08` Stop-limit (40=4) with StopPx and price **Info**
- ☐ `OT-095` Unsupported OrdType 40=5 **Blocker**
- ☐ `OT-099` Unsupported OrdType 40=9 **Blocker**

Validity (59)

- ☑ `TIF-0` Day (59=0) resting limit
- ☑ `TIF-1` GTC (59=1) resting limit
- ☑ `TIF-3a` IOC (59=3) partially fillable (50 of 100)
- ☑ `TIF-3b` IOC (59=3) with no liquidity
- ☑ `TIF-4a` FOK (59=4) fully fillable
- ☑ `TIF-4b` FOK (59=4) insufficient liquidity
- ☐ `TIF-6a` GTD (59=6) without expiry date (OMS sends none) **Gap**
- ☐ `TIF-6b` GTD (59=6) with ExpireTime (126) 3 s ahead — expires **Gap**
- ☑ `TIF-6c` GTD (59=6) with ExpireDate (432) only
- ☐ `TIF-2` At the Opening (59=2) **Gap**
- ☐ `TIF-5` Good Till Crossing (59=5) **Gap**
- ☐ `TIF-7` At the Close (59=7) **Gap**
- ☐ `TIF-S` Session (59=S) — offered by the OMS trader UI **Gap**
- ☑ `TIF-X` Missing TimeInForce

Side & quantity

- ☑ `SD-1` Buy (54=1)
- ☑ `SD-2` Sell (54=2)
- ☑ `SD-3` Missing Side (54)
- ☐ `SD-4` Side 54=5 (sell short) **Blocker**
- ☑ `QT-1` OrderQty = 0
- ☑ `QT-2` Fractional OrderQty (10.5)

- ☑ QT-3 Negative OrderQty
- ☑ QT-4 Quantity a multiple of 1000
- ☑ QT-5 Quantity 1500 on a lot-1000 instrument
- ☑ QT-5b Quantity 2000 on a lot-1000 instrument
- ☐ QT-6 Duplicate ClOrdID (11) resubmitted **Gap**
- ☑ QT-7 Missing Symbol (55)
- ☑ QT-8 Unknown symbol

Board group (762)

- ☑ BG-01 762=PUBLIC on a MAIN-board equity
- ☐ BG-02 762=ATBPUB on a MAIN-board equity **Gap**
- ☐ BG-03 762=SPUBLIC on a MAIN-board equity **Gap**
- ☐ BG-04 762=BLOCK on a MAIN-board equity **Gap**
- ☐ BG-05 762=SBLOCK on a MAIN-board equity **Gap**
- ☐ BG-06 762=BUYIN on a MAIN-board equity **Gap**
- ☐ BG-07 762=ATBBUYIN on a MAIN-board equity **Gap**
- ☐ BG-08 762=SBUYIN on a MAIN-board equity **Gap**
- ☐ BG-09 762=DEBT on a MAIN-board equity **Gap**
- ☐ BG-10 762=ATBDEBT on a MAIN-board equity **Gap**
- ☐ BG-11 762=SDEBT on a MAIN-board equity **Gap**
- ☐ BG-12 762=ALTDBT on a MAIN-board equity **Gap**
- ☐ BG-13 762=BUYDBT on a MAIN-board equity **Gap**
- ☐ BG-14 762=YIELDDBT on a MAIN-board equity **Gap**
- ☐ BG-15 762=INDEX on a MAIN-board equity **Gap**
- ☐ BG-16 762=ATBX on a MAIN-board equity **Gap**
- ☐ BG-17 762=SCPX on a MAIN-board equity **Gap**
- ☑ BG-18 762=SPUBLIC on an SME-board equity
- ☑ BG-19 762=ATBPUB on an ATB-board equity
- ☑ BG-20 Missing 762

ExecInst (18)

- ☐ EI-1 18=Placed (n) on a new order **Partial**
- ☐ EI-2 18=Park (S) on a new order **Gap**
- ☐ EI-3 18=Block (w) on a new order **Gap**
- ☐ EI-4 18=Block Park (w S) on a new order **Gap**
- ☐ EI-5 18=q (Unpark) on a modify (35=G) **Partial**
- ☐ EI-6 18=w q (Unpark) on a modify (35=G) **Gap**

Quantity options

- ☐ QO-1 DisplayQty (1138=30) iceberg on a 100-lot (the OMS's tag) **Gap**
- ☐ QO-2 MaxFloor (111=30) iceberg (engine's own tag) **Info**
- ☐ QO-3 MinQty (110) greater than OrderQty (500 > 100) **Gap**
- ☐ QO-4 MinQty (110=50) with only 30 available **Gap**

Tags & reports

- ☑ TG-1 Party group 453 (448/447/452) present
- ☑ TG-2 No party group
- ☑ TG-3 OrderRestrictions 529=N (non-standard)
- ☐ TG-4 Missing Account (1) **Info**
- ☑ TG-5 Missing TransactTime (60)
- ☑ TG-6 Missing ClOrdID (11)
- ☑ ER-1 Fill ER carries FIX-required tags (37,17,150,39,55,54,151,14,11)
- ☐ ER-2 Fill ER ExecType (150) **Blocker**
- ☐ ER-3 ER echoes Account (1) **Gap**
- ☐ ER-4 ER echoes OrdType (40) / TimeInForce (59) **Gap**
- ☑ ER-5 ER carries last fill price/qty (31/32)
- ☑ ER-6 ER carries average price (6) and cumulative qty (14)
- ☑ ER-7 ER carries security id (48/22) and board
- ☐ ER-8 Reject ER carries OrdRejReason (103) and Text (58) **Partial**

Price band / tick

- ☑ All circuit-limit tier boundaries (64/64 probes correct)
- ☑ PT-1 Price on tick (100.10)
- ☑ PT-2 Price off tick (100.05)
- ☑ PT-3 Price 100.02 (sub-paisa tick)
- ☐ PT-4 Market order ignores the price band **Info**
- ☑ PT-5 Order on a halted instrument
- ☑ PT-6 Order after the halt is lifted

Cancel / modify / status

- ☑ CM-1 Cancel a resting order (35=F)
- ☑ CM-2 Cancel an unknown order
- ☐ CM-3 35=9 carries OrderID (37) — required by FIX **Blocker**
- ☑ CM-4 Modify price WITHOUT Symbol/762 (as the OMS sends 35=G)
- ☑ CM-5 Modify quantity 100 -> 60
- ☑ CM-6 Modify an unknown order
- ☐ CM-7 Modify to a price off tick (96.05) **Gap**

- ☐ CM-8 Modify to a price outside the band (150.00) Gap
- ☑ CM-9 Modify using the latest ClOrdID chain (replace a replaced order)
- ☐ CM-10 Replace ER echoes OrigClOrdID (41) Gap
- ☐ CM-11 Order Status Request (35=H) on a live order Gap
- ☐ CM-12 Order Status Request (35=H) on an unknown order Gap
- ☐ CM-13 Modify a parked stop order Info

Phases

- ☑ PH-1 Limit order during Pre-Open
- ☑ PH-2 Market order during Pre-Open
- ☐ PH-3 Limit far outside the band during Pre-Open Gap
- ☐ PH-4 Uncross at the Pre-Open -> Continuous transition Gap
- ☑ PH-5 GTC order survives a phase change

Matching

- ☑ MT-1 Buy limit crosses the best ask — price improvement
- ☑ MT-2 Price-time priority (two asks at the same price)
- ☑ MT-3 Partial fill then rest (day limit)

Rules not modelled

- ☐ ST-1 Day orders expire at market close Gap
- ☐ ST-2 Post-close session: Market order priced at the close price Gap
- ☐ ST-3 Automatic circuit breaker / volatility halt Gap
- ☐ ST-4 Category-based rules (A/B/G/N/Z) and board-specific rules Gap
- ☐ ST-5 Pre-trade RMS (buying power, holdings, position limit) on the FIX path Info
- ☐ ST-6 Order-status / execution-report replay on reconnect Info

Session messages

- ☑ SS-1 Logon (35=A) with ResetSeqNumFlag 141=Y
- ☑ SS-2 TestRequest (35=1, 112=PING)
- ☑ SS-3 ResendRequest (35=2, 7=1, 16=0)
- ☑ SS-4 SequenceReset (35=4, 123=N, 36=5000)
- ☑ SS-5 Logout (35=5)
- ☑ SS-6 Heartbeat (35=0) interval 30 s

Evidence — sample messages for each deviation

SOH delimiters shown as "|". The sent string is the OMS-style message as it left the test initiator; session header fields are added by QuickFIX.

OT-02 — Limit (40=2) without price (44), liquid book Deviates

SENT

```
8=FIXT.1.1|9=190|35=D|34=3|49=TESTBRK|50=TRADER1|52=20261013-06:40:58.121|56=DSEME|1=BO1001|11=CF00002|38=100|
40=2|54=1|55=T002M|59=0|60=20261013-06:40:58.121|453=1|448=CLIENT01|447=C|452=5|529=N|762=PUBLIC|10=205|
```

RECEIVED

```
8=FIXT.1.1|9=216|35=8|34=3|49=DSEME|52=20261013-06:40:58.121|56=TESTBRK|6=100.00|11=CF00002|14=100|17=2|22=4|
31=100.00|32=100|37=3|38=100|39=2|48=BD0001000002|54=1|55=T002M|60=20261013-06:40:58.121|150=TRADE|151=0|
207=XDHA|1300=MAIN|10=089|
```

accepted and executed — the engine silently converts a 40=2 order with no price into a MARKET order (fixacceptor.go:465-479)

OT-06 — Market-At-Best (40=Z) Gap

SENT

```
8=FIXT.1.1|9=190|35=D|34=7|49=TESTBRK|50=TRADER1|52=20261013-06:40:58.603|56=DSEME|1=BO1001|11=CF00006|38=100|
40=Z|54=1|55=T004M|59=0|60=20261013-06:40:58.603|453=1|448=CLIENT01|447=C|452=5|529=N|762=PUBLIC|10=009|
```

RECEIVED

```
8=FIXT.1.1|9=140|35=3|34=7|49=DSEME|52=20261013-06:40:58.604|56=TESTBRK|57=TRADER1|45=7|58=Value is incorrect
(out of range) for this tag|371=40|372=D|373=5|10=222|
```

OrdType Z is not in the engine's data dictionary, so the session layer refuses it (35=3) — Market-At-Best is not supported

OT-095 — Unsupported OrdType 40=5 Deviates

SENT

```
8=FIXT.1.1|9=201|35=D|34=10|49=TESTBRK|50=TRADER1|52=20261013-06:40:58.966|56=DSEME|1=BO1001|11=CF00009|
38=100|40=5|44=100.00|54=1|55=T004M|59=0|60=20261013-06:40:58.966|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=231|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=10|49=DSEME|52=20261013-06:40:58.966|56=TESTBRK|6=0.00|11=CF00009|14=0|17=8|22=4|
37=10|38=100|39=0|44=100.00|48=BD0001000004|54=1|55=T004M|60=20261013-06:40:58.966|150=0|151=100|207=XDHA|
1300=MAIN|10=019|
```

accepted and silently treated as a Limit order (mapping default in fixacceptor.go:465-479)

OT-099 — Unsupported OrdType 40=9 Deviates

SENT

```
8=FIXT.1.1|9=201|35=D|34=11|49=TESTBRK|50=TRADER1|52=20261013-06:40:59.086|56=DSEME|1=BO1001|11=CF00010|
38=100|40=9|44=100.00|54=1|55=T004M|59=0|60=20261013-06:40:59.086|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=216|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=11|49=DSEME|52=20261013-06:40:59.086|56=TESTBRK|6=0.00|11=CF00010|14=0|17=9|22=4|
37=11|38=100|39=0|44=100.00|48=BD0001000004|54=1|55=T004M|60=20261013-06:40:59.086|150=0|151=100|207=XDHA|
1300=MAIN|10=002|
```

accepted and silently treated as a Limit order (mapping default in fixacceptor.go:465-479)

TIF-6a — GTD (59=6) without expiry date (OMS sends none) Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=18|49=TESTBRK|50=TRADER1|52=20261013-06:40:59.931|56=DSEME|1=BO1001|11=CF00017|
38=100|40=2|44=95.00|54=1|55=T011M|59=6|60=20261013-06:40:59.931|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=189|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=18|49=DSEME|52=20261013-06:40:59.931|56=TESTBRK|6=0.00|11=CF00017|14=0|17=16|22=4|
37=21|38=100|39=0|44=95.00|48=BD0001000011|54=1|55=T011M|60=20261013-06:40:59.931|150=0|151=100|207=XDHA|
1300=MAIN|10=022|
```

accepted; with no expiry the order never expires (parseExpire returns zero)

TIF-6b — GTD (59=6) with ExpireTime (126) 3 s ahead — expires Gap

SENT

```
8=FIXT.1.1|9=226|35=D|34=19|49=TESTBRK|50=TRADER1|52=20261013-06:41:00.052|56=DSEME|1=BO1001|11=CF00018|
38=100|40=2|44=94.00|54=1|55=T011M|59=6|60=20261013-06:41:00.052|126=20261013-06:41:03.052|453=1|448=CLIENT01|
447=C|452=5|529=N|762=PUBLIC|10=154|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=19|49=DSEME|52=20261013-06:41:00.052|56=TESTBRK|6=0.00|11=CF00018|14=0|17=17|22=4|
37=22|38=100|39=0|44=94.00|48=BD0001000011|54=1|55=T011M|60=20261013-06:41:00.052|150=0|151=100|207=XDHA|
1300=MAIN|10=243|
```

order removed from the book at expiry but NO Execution Report (150=C) was sent — the broker is never told

TIF-2 — At the Opening (59=2) Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=21|49=TESTBRK|50=TRADER1|52=20261013-06:41:05.095|56=DSEME|1=BO1001|11=CF00020|
38=100|40=2|44=92.00|54=1|55=T011M|59=2|60=20261013-06:41:05.095|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=156|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=21|49=DSEME|52=20261013-06:41:05.095|56=TESTBRK|6=0.00|11=CF00020|14=0|17=19|22=4|
37=24|38=100|39=0|44=92.00|48=BD0001000011|54=1|55=T011M|60=20261013-06:41:05.095|150=0|151=100|207=XDHA|
1300=MAIN|10=255|
```

accepted but silently treated as Day (mapTIF default) — semantics not implemented

TIF-5 — Good Till Crossing (59=5) Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=22|49=TESTBRK|50=TRADER1|52=20261013-06:41:05.216|56=DSEME|1=BO1001|11=CF00021|
38=100|40=2|44=92.00|54=1|55=T011M|59=5|60=20261013-06:41:05.216|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=151|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=22|49=DSEME|52=20261013-06:41:05.216|56=TESTBRK|6=0.00|11=CF00021|14=0|17=20|22=4|
37=25|38=100|39=0|44=92.00|48=BD0001000011|54=1|55=T011M|60=20261013-06:41:05.216|150=0|151=100|207=XDHA|
1300=MAIN|10=240|
```

accepted but silently treated as Day (mapTIF default) — semantics not implemented

TIF-7 — At the Close (59=7) Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=23|49=TESTBRK|50=TRADER1|52=20261013-06:41:05.337|56=DSEME|1=BO1001|11=CF00022|
38=100|40=2|44=92.00|54=1|55=T011M|59=7|60=20261013-06:41:05.337|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=163|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=23|49=DSEME|52=20261013-06:41:05.337|56=TESTBRK|6=0.00|11=CF00022|14=0|17=21|22=4|
37=26|38=100|39=0|44=92.00|48=BD0001000011|54=1|55=T011M|60=20261013-06:41:05.337|150=0|151=100|207=XDHA|
1300=MAIN|10=252|
```

accepted but silently treated as Day (mapTIF default) — semantics not implemented

TIF-S — Session (59=S) — offered by the OMS trader UI Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=24|49=TESTBRK|50=TRADER1|52=20261013-06:41:05.458|56=DSEME|1=BO1001|11=CF00023|
38=100|40=2|44=91.00|54=1|55=T011M|59=S|60=20261013-06:41:05.458|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=200|
```

RECEIVED

```
8=FIXT.1.1|9=142|35=3|34=24|49=DSEME|52=20261013-06:41:05.458|56=TESTBRK|57=TRADER1|45=24|58=Value is
incorrect (out of range) for this tag|371=59|372=D|373=5|10=072|
```

rejected by the strict dictionary (35=3) — an order the OMS UI allows can never reach the engine

SD-4 — Side 54=5 (sell short) Deviates

SENT

```
8=FIXT.1.1|9=201|35=D|34=29|49=TESTBRK|50=TRADER1|52=20261013-06:41:06.061|56=DSEME|1=BO1001|11=CF00028|
38=100|40=2|44=100.00|54=5|55=T012M|59=0|60=20261013-06:41:06.061|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=202|
```

RECEIVED

```
8=FIXT.1.1|9=206|35=8|34=29|49=DSEME|52=20261013-06:41:06.062|56=TESTBRK|6=0.00|11=CF00028|14=0|17=25|22=4|
37=30|38=100|39=0|44=100.00|48=BD0001000012|54=1|55=T012M|60=20261013-06:41:06.062|150=0|151=100|207=XDHA|
1300=MAIN|10=040|
```

accepted — any Side other than '2' is treated as a BUY (fixacceptor.go:461-464), so a sell-short would execute as a purchase

QT-6 — Duplicate ClOrdID (11) resubmitted Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=37|49=TESTBRK|50=TRADER1|52=20261013-06:41:07.028|56=DSEME|1=BO1001|11=CF00035|
38=100|40=2|44=95.00|54=1|55=T012M|59=0|60=20261013-06:41:06.906|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=171|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=37|49=DSEME|52=20261013-06:41:07.028|56=TESTBRK|6=0.00|11=CF00035|14=0|17=33|22=4|
37=34|38=100|39=0|44=95.00|48=BD0001000012|54=1|55=T012M|60=20261013-06:41:07.028|150=0|151=100|207=XDHA|
1300=MAIN|10=010|
```

second order with the same ClOrdID accepted — no duplicate-ClOrdID check (also breaks cancel/replace by ClOrdID)

BG-02 — 762=ATBPUB on a MAIN-board equity Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=41|49=TESTBRK|50=TRADER1|52=20261013-06:41:07.511|56=DSEME|1=BO1001|11=CF00039|
38=100|40=2|44=95.00|54=1|55=T013M|59=0|60=20261013-06:41:07.511|453=1|448=CLIENT01|447=C|452=5|529=N|
762=ATBPUB|10=160|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=41|49=DSEME|52=20261013-06:41:07.511|56=TESTBRK|6=0.00|11=CF00039|14=0|17=37|22=4|
37=36|38=100|39=0|44=95.00|48=BD0001000013|54=1|55=T013M|60=20261013-06:41:07.511|150=0|151=100|207=XDHA|
1300=MAIN|10=011|
```

accepted — tag 762 is never read, so board groups are not validated against the instrument

EI-2 — 18=Park (S) on a new order Gap

SENT

```
8=FIXT.1.1|9=205|35=D|34=61|49=TESTBRK|50=TRADER1|52=20261013-06:41:09.929|56=DSEME|1=BO1001|11=CF00059|18=S|
38=100|40=2|44=95.00|54=1|55=T017M|59=0|60=20261013-06:41:09.929|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=198|
```

RECEIVED

```
8=FIXT.1.1|9=205|35=8|34=61|49=DSEME|52=20261013-06:41:09.929|56=TESTBRK|6=0.00|11=CF00059|14=0|17=57|22=4|
37=56|38=100|39=0|44=95.00|48=BD0001000017|54=1|55=T017M|60=20261013-06:41:09.929|150=0|151=100|207=XDHA|
1300=MAIN|10=057|
```

accepted but the order rests on the book — Park is not implemented (tag 18 never read)

EI-3 — 18=Block (w) on a new order Gap

SENT

```
8=FIXT.1.1|9=205|35=D|34=62|49=TESTBRK|50=TRADER1|52=20261013-06:41:10.050|56=DSEME|1=BO1001|11=CF00060|18=w|
38=100|40=2|44=95.00|54=1|55=T018M|59=0|60=20261013-06:41:10.049|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=190|
```

RECEIVED

```
8=FIXT.1.1|9=142|35=3|34=62|49=DSEME|52=20261013-06:41:10.050|56=TESTBRK|57=TRADER1|45=62|58=Value is
incorrect (out of range) for this tag|371=18|372=D|373=5|10=055|
```

rejected — 'w' is not in the engine's tag-18 dictionary enum, so the OMS's DSE-specific instruction never reaches the engine

EI-4 — 18=Block Park (w S) on a new order Gap

SENT

8=FIXT.1.1|9=207|35=D|34=63|49=TESTBRK|50=TRADER1|52=20261013-06:41:10.171|56=DSEME|1=BO1001|11=CF00061|18=w
S|38=100|40=2|44=95.00|54=1|55=T019M|59=0|60=20261013-06:41:10.171|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=054|

RECEIVED

8=FIXT.1.1|9=142|35=3|34=63|49=DSEME|52=20261013-06:41:10.171|56=TESTBRK|57=TRADER1|45=63|58=Value is
incorrect (out of range) for this tag|371=18|372=D|373=5|10=061|

rejected — 'w S' is not in the engine's tag-18 dictionary enum, so the OMS's DSE-specific instruction never reaches the engine

EI-6 — 18=w q (Unpark) on a modify (35=G) Gap

SENT

8=FIXT.1.1|9=198|35=G|34=67|49=TESTBRK|50=TRADER1|52=20261013-06:41:10.653|56=DSEME|1=BO1001|11=CF00065|18=w
q|38=100|40=2|41=CF00064|44=95.00|54=1|59=0|60=20261013-06:41:10.653|453=1|448=CLIENT01|447=C|452=5|529=N|
10=025|

RECEIVED

8=FIXT.1.1|9=142|35=3|34=67|49=DSEME|52=20261013-06:41:10.653|56=TESTBRK|57=TRADER1|45=67|58=Value is
incorrect (out of range) for this tag|371=18|372=G|373=5|10=077|

rejected — Unpark ('w q') cannot be sent through the dictionary-validated session

QO-1 — DisplayQty (1138=30) iceberg on a 100-lot (the OMS's tag) Gap

SENT

8=FIXT.1.1|9=208|35=D|34=68|49=TESTBRK|50=TRADER1|52=20261013-06:41:10.774|56=DSEME|1=BO1001|11=CF00066|
38=100|40=2|44=90.00|54=1|55=T021M|59=0|60=20261013-06:41:10.774|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|1138=30|10=036|

RECEIVED

8=FIXT.1.1|9=205|35=8|34=68|49=DSEME|52=20261013-06:41:10.774|56=TESTBRK|6=0.00|11=CF00066|14=0|17=61|22=4|
37=59|38=100|39=0|44=90.00|48=BD0001000021|54=1|55=T021M|60=20261013-06:41:10.774|150=0|151=100|207=XDHA|
1300=MAIN|10=025|

book shows 100 — tag 1138 is ignored (the engine reads tag 111 MaxFloor instead), so the iceberg is not applied to OMS orders

QO-3 — MinQty (110) greater than OrderQty (500 > 100) Gap

SENT

8=FIXT.1.1|9=208|35=D|34=70|49=TESTBRK|50=TRADER1|52=20261013-06:41:11.016|56=DSEME|1=BO1001|11=CF00068|
38=100|40=2|44=90.00|54=1|55=T023M|59=0|60=20261013-06:41:11.016|110=500|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=004|

RECEIVED

8=FIXT.1.1|9=205|35=8|34=70|49=DSEME|52=20261013-06:41:11.016|56=TESTBRK|6=0.00|11=CF00068|14=0|17=63|22=4|
37=61|38=100|39=0|44=90.00|48=BD0001000023|54=1|55=T023M|60=20261013-06:41:11.016|150=0|151=100|207=XDHA|
1300=MAIN|10=255|

accepted — MinQty (110) is ignored

QO-4 — MinQty (110=50) with only 30 available Gap

SENT

8=FIXT.1.1|9=208|35=D|34=71|49=TESTBRK|50=TRADER1|52=20261013-06:41:11.137|56=DSEME|1=BO1001|11=CF00069|
38=100|40=2|44=100.00|54=1|55=T024M|59=0|60=20261013-06:41:11.137|110=50|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=007|

RECEIVED

8=FIXT.1.1|9=228|35=8|34=71|49=DSEME|52=20261013-06:41:11.137|56=TESTBRK|6=100.00|11=CF00069|14=30|17=64|22=4|
31=100.00|32=30|37=63|38=100|39=1|44=100.00|48=BD0001000024|54=1|55=T024M|60=20261013-06:41:11.137|150=TRADE|
151=70|207=XDHA|1300=MAIN|10=173|

filled below the minimum quantity — MinQty (110) is ignored

ER-2 — Fill ER ExecType (150) Deviates

SENT

```
8=FIXT.1.1|9=201|35=D|34=78|49=TESTBRK|50=TRADER1|52=20261013-06:41:11.984|56=DSEME|1=BO1001|11=CF00076|
38=100|40=2|44=100.00|54=1|55=T026M|59=0|60=20261013-06:41:11.984|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=230|
```

RECEIVED

```
8=FIXT.1.1|9=229|35=8|34=78|49=DSEME|52=20261013-06:41:11.984|56=TESTBRK|6=100.00|11=CF00076|14=100|17=69|
22=4|31=100.00|32=100|37=69|38=100|39=2|44=100.00|48=BD0001000026|54=1|55=T026M|60=20261013-06:41:11.984|
150=TRADE|151=0|207=XDHA|1300=MAIN|10=252|
```

150=TRADE — not a valid FIX 5.0 value (must be 'F')

ER-3 — ER echoes Account (1) Gap

SENT

```
8=FIXT.1.1|9=201|35=D|34=78|49=TESTBRK|50=TRADER1|52=20261013-06:41:11.984|56=DSEME|1=BO1001|11=CF00076|
38=100|40=2|44=100.00|54=1|55=T026M|59=0|60=20261013-06:41:11.984|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=230|
```

RECEIVED

```
8=FIXT.1.1|9=229|35=8|34=78|49=DSEME|52=20261013-06:41:11.984|56=TESTBRK|6=100.00|11=CF00076|14=100|17=69|
22=4|31=100.00|32=100|37=69|38=100|39=2|44=100.00|48=BD0001000026|54=1|55=T026M|60=20261013-06:41:11.984|
150=TRADE|151=0|207=XDHA|1300=MAIN|10=252|
```

tag 1 not echoed on the execution report

ER-4 — ER echoes OrdType (40) / TimeInForce (59) Gap

SENT

```
8=FIXT.1.1|9=201|35=D|34=78|49=TESTBRK|50=TRADER1|52=20261013-06:41:11.984|56=DSEME|1=BO1001|11=CF00076|
38=100|40=2|44=100.00|54=1|55=T026M|59=0|60=20261013-06:41:11.984|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=230|
```

RECEIVED

```
8=FIXT.1.1|9=229|35=8|34=78|49=DSEME|52=20261013-06:41:11.984|56=TESTBRK|6=100.00|11=CF00076|14=100|17=69|
22=4|31=100.00|32=100|37=69|38=100|39=2|44=100.00|48=BD0001000026|54=1|55=T026M|60=20261013-06:41:11.984|
150=TRADE|151=0|207=XDHA|1300=MAIN|10=252|
```

tags 40/59 not echoed

CM-3 — 35=9 carries OrderID (37) — required by FIX Deviates

SENT

```
8=FIXT.1.1|9=145|35=F|34=154|49=TESTBRK|50=TRADER1|52=20261013-06:41:21.176|56=DSEME|1=BO1001|11=CF00152|
38=100|41=NOSUCH2|54=1|55=T046M|60=20261013-06:41:21.176|10=060|
```

RECEIVED

```
8=FIXT.1.1|9=129|35=9|34=154|49=DSEME|52=20261013-06:41:21.176|56=TESTBRK|11=CF00152|39=8|41=NOSUCH2|
58=unknown or already-done order|102=1|434=1|10=209|
```

35=9 has no OrderID (37) — fixacceptor.go:531-543

CM-7 — Modify to a price off tick (96.05) Gap

SENT

```
8=FIXT.1.1|9=192|35=G|34=161|49=TESTBRK|50=TRADER1|52=20261013-06:41:22.022|56=DSEME|1=BO1001|11=CF00159|
38=100|40=2|41=CF00158|44=96.05|54=1|59=0|60=20261013-06:41:22.022|453=1|448=CLIENT01|447=C|452=5|529=N|
10=143|
```

RECEIVED

```
8=FIXT.1.1|9=168|35=8|34=161|49=DSEME|52=20261013-06:41:22.022|56=TESTBRK|6=0.00|11=CF00159|14=0|17=149|
37=144|38=100|39=0|54=1|55=T046M|60=20261013-06:41:22.022|150=5|151=100|207=XDHA|10=248|
```

accepted — replace does not re-run the tick check

CM-8 — Modify to a price outside the band (150.00) Gap

SENT

```
8=FIXT.1.1|9=193|35=G|34=163|49=TESTBRK|50=TRADER1|52=20261013-06:41:22.264|56=DSEME|1=BO1001|11=CF00161|
38=100|40=2|41=CF00160|44=150.00|54=1|59=0|60=20261013-06:41:22.264|453=1|448=CLIENT01|447=C|452=5|529=N|
10=182|
```

RECEIVED

```
8=FIXT.1.1|9=168|35=8|34=163|49=DSEME|52=20261013-06:41:22.264|56=TESTBRK|6=0.00|11=CF00161|14=0|17=151|
37=145|38=100|39=0|54=1|55=T046M|60=20261013-06:41:22.264|150=5|151=100|207=XDHA|10=253|
```

accepted — replace does not re-run the price-band check, so an order can be amended to any price

CM-10 — Replace ER echoes OrigClOrdID (41) Gap

SENT

```
8=FIXT.1.1|9=192|35=G|34=168|49=TESTBRK|50=TRADER1|52=20261013-06:41:22.868|56=DSEME|1=BO1001|11=CF00166|
38=100|40=2|41=CF00165|44=95.50|54=1|59=0|60=20261013-06:41:22.868|453=1|448=CLIENT01|447=C|452=5|529=N|
10=181|
```

RECEIVED

```
8=FIXT.1.1|9=168|35=8|34=168|49=DSEME|52=20261013-06:41:22.868|56=TESTBRK|6=0.00|11=CF00166|14=0|17=156|
37=147|38=100|39=0|54=1|55=T046M|60=20261013-06:41:22.868|150=5|151=100|207=XDHA|10=034|
```

OrigClOrdID (41) not echoed on the execution report

CM-11 — Order Status Request (35=H) on a live order Gap

SENT

```
8=FIXT.1.1|9=101|35=H|34=170|49=TESTBRK|50=TRADER1|52=20261013-06:41:23.109|56=DSEME|11=CF00167|54=1|55=T046M|
790=ST1|10=127|
```

RECEIVED

```
(no response)
```

no reply — 35=H is not handled (fixacceptor.go:403-410); the OMS cannot reconcile order state

CM-12 — Order Status Request (35=H) on an unknown order Gap

SENT

```
8=FIXT.1.1|9=101|35=H|34=171|49=TESTBRK|50=TRADER1|52=20261013-06:41:23.809|56=DSEME|11=NOSUCH4|54=1|55=T046M|
790=ST2|10=005|
```

RECEIVED

```
(no response)
```

no reply — 35=H is not handled

PH-3 — Limit far outside the band during Pre-Open Gap

SENT

```
8=FIXT.1.1|9=202|35=D|34=176|49=TESTBRK|50=TRADER1|52=20261013-06:41:24.994|56=DSEME|1=BO1001|11=CF00172|
38=100|40=2|44=150.00|54=1|55=T049M|59=0|60=20261013-06:41:24.994|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=039|
```

RECEIVED

```
8=FIXT.1.1|9=178|35=8|34=174|49=DSEME|52=20261013-06:41:24.994|56=TESTBRK|6=0.00|11=CF00172|14=0|17=161|
37=153|38=100|39=0|44=150.00|54=1|55=T049M|60=20261013-06:41:24.994|150=0|151=100|207=XDHA|10=226|
```

accepted — Pre-Open skips the tick and price-band checks (engine.go:278-289 returns before them)

PH-4 — Uncross at the Pre-Open -> Continuous transition Gap

SENT

```
(phase change)
```

RECEIVED

```
(no response)
```

the crossed order was executed in the uncross but the broker received NO Execution Report — auction fills are not reported over FIX

Recommended fixes (not applied)

#	Priority	Change	Where
1	Deviates	**Reject unsupported OrdType and Side; require a price on 40=2** fixacceptor.go:461-479	Map only 1/2/3/4 (+Z once supported); reject everything else and any 40=2 without a price; accept only Side 1/2.
2	Deviates	**Send ExecType F for fills; add OrderID to 35=9; set OrdRejReason (103)** fixacceptor.go:531-543, 587-648	Change the literal 'TRADE' to 'F'. Echo tags 1, 41, 40, 59 on reports.
3	Gap	**Handle Order Status Request (35=H)** fixacceptor.go:403-411	Return an Execution Report with ExecType I and the current OrdStatus; reject unknown orders.

#	Priority	Change	Where
4	Gap	**Re-run tick and price-band checks on replace; apply them in Pre-Open** engine.go:278-329, 595-625	Move the pre-trade block ahead of the Pre-Open branch and call it from doReplace.
6	Gap	**Support Market-At-Best (Z), Validity S, ExecInst n/S/q/w/'w S'/'w q'** spec/FIX50SP2.xml, fixacceptor.go	Extend the data dictionary enums and implement Park/Unpark/Block semantics on the book.
7	Gap	**DisplayQty (1138) and MinQty (110)** fixacceptor.go:458, engine.go:502	Read 1138 as the iceberg peak (keep 111 as an alias); enforce MinQty on matching and reject MinQty > OrderQty.
8	Gap	**Validate tag 762 against the instrument's board/category** fixacceptor.go validateNew	Add board-group ↔ board mapping; reject Index groups for orders.
9	Gap	**GTD: require expiry, send an Execution Report on expiry; implement Day expiry at close** fixacceptor.go:564-578, engine.go:765-781	Reject GTD without 126/432; emit ExecType C when the sweeper removes an order; add an end-of-day purge.
10	Gap	**Send Execution Reports for auction (uncross) fills; add post-close session and automatic circuit halts** engine.go:627-725	Route uncross trades through the order's notify callback; add a post-close phase with close-price market orders.
11	Gap	**Reject duplicate ClOrdID** engine.go (process)	A repeated 11 currently creates a second order and shadows the first for cancel/replace.

Out of scope for this report: OMS-side defects listed in the OMS reference (GTD sent without expiry, modify without Symbol/762, Market-At-Best blocked by the OMS permission check) — they are on the sending side. Items DSE has never been observed to accept or reject (block, buy-in, iceberg, GTD, cancel, modify, status) are judged against the FIX standard and the OMS reference only and should be re-confirmed against DSE.